

- Create a vpc, create public subnet, launch ec2 instance inside the vpc, security group,
- Security group should allow ssh and 443 access for that instance and map that security group to the EC2 instance.
- Variabilize resources, define variables in variables.tf and values in the custom.tfvars file.

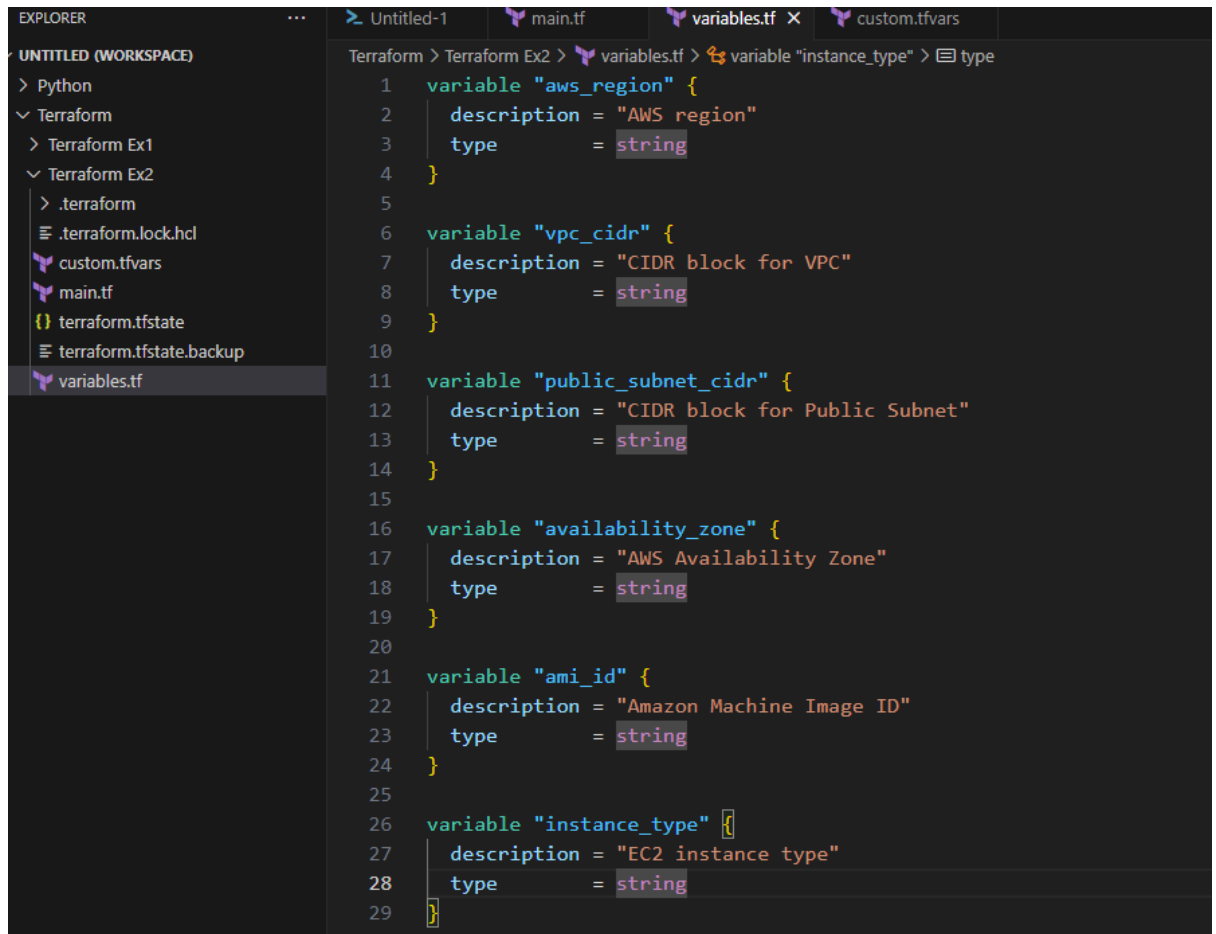
1. Setup the access key and secret key as environment variables inside the powershell.

```
$env:AWS_ACCESS_KEY_ID="AKIAVFIXXXXXXXXXXXXX"

$env:AWS_SECRET_ACCESS_KEY="5+O8GnIFC0xxxxxxxxnxxxxxxxxess"

$env:AWS_REGION="us-east-2"
```

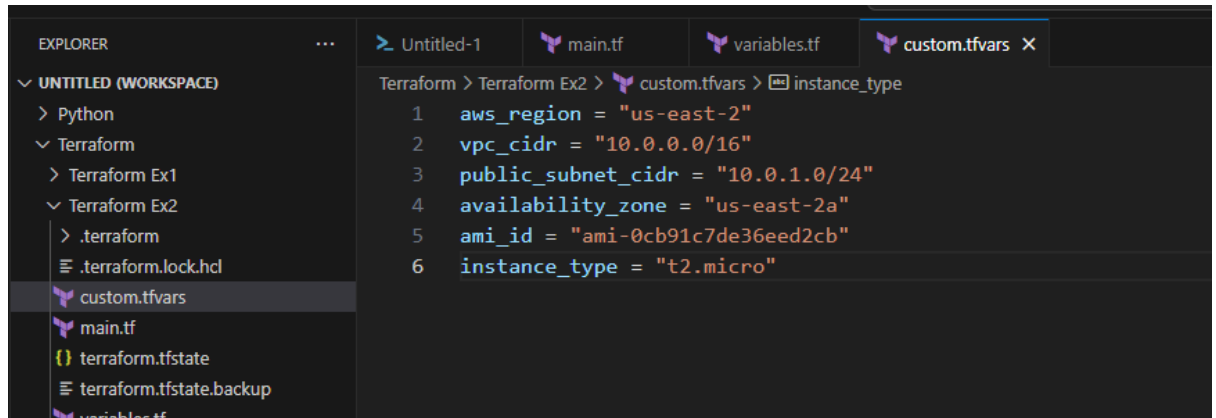
2. Create variables.tf and add variable declarations.



The screenshot shows the VS Code interface with the Explorer on the left and the Terraform workspace open. The Explorer shows a project structure for Terraform Ex2, including .terraform, custom.tfvars, main.tf, terraform.tfstate, and variables.tf. The main.tf file is open, showing Terraform configuration for an EC2 instance. The variables.tf file is also open, showing variable declarations for aws_region, vpc_cidr, public_subnet_cidr, availability_zone, ami_id, and instance_type.

```
1  variable "aws_region" {
2    description = "AWS region"
3    type       = string
4  }
5
6  variable "vpc_cidr" {
7    description = "CIDR block for VPC"
8    type       = string
9  }
10
11 variable "public_subnet_cidr" {
12   description = "CIDR block for Public Subnet"
13   type       = string
14 }
15
16 variable "availability_zone" {
17   description = "AWS Availability Zone"
18   type       = string
19 }
20
21 variable "ami_id" {
22   description = "Amazon Machine Image ID"
23   type       = string
24 }
25
26 variable "instance_type" {
27   description = "EC2 instance type"
28   type       = string
29 }
```

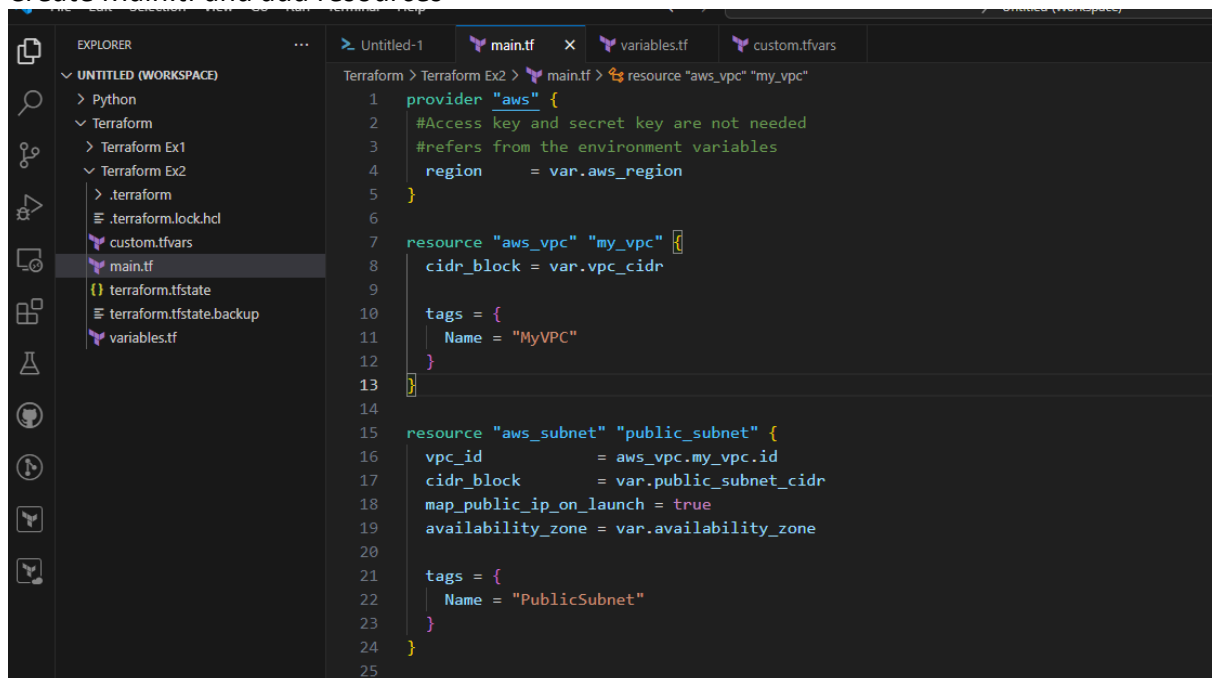
3. Create custom.tfvars file and initialize values to variables



The screenshot shows the VS Code interface with the Explorer on the left and the Editor on the right. The Explorer shows a workspace with folders 'Python' and 'Terraform'. Under 'Terraform', there are subfolders 'Terraform Ex1' and 'Terraform Ex2'. Inside 'Terraform Ex2', there are files '.terraform', '.terraform.lock.hcl', 'custom.tfvars', 'main.tf', 'terraform.tfstate', 'terraform.tfstate.backup', and 'variables.tf'. The 'custom.tfvars' file is selected. The Editor shows the content of 'custom.tfvars' with the following code:

```
Terraform > Terraform Ex2 > custom.tfvars > instance_type
1  aws_region = "us-east-2"
2  vpc_cidr = "10.0.0.0/16"
3  public_subnet_cidr = "10.0.1.0/24"
4  availability_zone = "us-east-2a"
5  ami_id = "ami-0cb91c7de36eed2cb"
6  instance_type = "t2.micro"
```

4. Create Main.tf and add resources



The screenshot shows the VS Code interface with the Explorer on the left and the Editor on the right. The Explorer shows the same workspace as the previous screenshot. The 'main.tf' file is selected. The Editor shows the content of 'main.tf' with the following code:

```
Terraform > Terraform Ex2 > main.tf > resource "aws_vpc" "my_vpc"
1  provider "aws" {
2    #Access key and secret key are not needed
3    #refers from the environment variables
4    region = var.aws_region
5  }
6
7  resource "aws_vpc" "my_vpc" {
8    cidr_block = var.vpc_cidr
9
10   tags = {
11     Name = "MyVPC"
12   }
13 }
14
15 resource "aws_subnet" "public_subnet" {
16   vpc_id = aws_vpc.my_vpc.id
17   cidr_block = var.public_subnet_cidr
18   map_public_ip_on_launch = true
19   availability_zone = var.availability_zone
20
21   tags = {
22     Name = "PublicSubnet"
23   }
24 }
25 }
```

Code: Main.tf

```
provider "aws" {
  #Access key and secret key are not needed
  #refers from the environment variables
  region = var.aws_region
}

resource "aws_vpc" "my_vpc" {
  cidr_block = var.vpc_cidr

  tags = {
    Name = "MyVPC"
  }
}

resource "aws_subnet" "public_subnet" {
  vpc_id            = aws_vpc.my_vpc.id
  cidr_block        = var.public_subnet_cidr
  map_public_ip_on_launch = true
  availability_zone = var.availability_zone

  tags = {
    Name = "PublicSubnet"
  }
}

resource "aws_security_group" "web_sg" {
  vpc_id = aws_vpc.my_vpc.id
  # Allow SSH (port 22) from anywhere
  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  # Allow HTTPS (port 443) from anywhere
  ingress {
    from_port = 443
    to_port   = 443
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  # Allow all outbound traffic
  egress {
    from_port = 0
    to_port   = 0
    protocol  = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
  tags = {
    Name = "WebSecurityGroup"
  }
}

resource "aws_instance" "web_server" {
  ami            = var.ami_id
  instance_type  = var.instance_type
  subnet_id      = aws_subnet.public_subnet.id
  vpc_security_group_ids = [aws_security_group.web_sg.id]

  tags = {
    Name = "NewWebServer"
  }
}
```

5. Terraform Output

```
PS C:\devops\Terraform\Terraform Ex2> terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v5.89.0
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
PS C:\devops\Terraform\Terraform Ex2> terraform plan -var-file="custom.tfvars"
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# aws_instance.web_server will be created
+ resource "aws_instance" "web_server" {
+   ami                         = "ami-0cb91c7de36eed2cb"
```

```
PS C:\devops\Terraform\Terraform Ex2> terraform apply -var-file="custom.tfvars"
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# aws_instance.web_server will be created
+ resource "aws_instance" "web_server" {
+   ami                         = "ami-0cb91c7de36eed2cb"
+   arn                        = (known after apply)
+   associate_public_ip_address = (known after apply)
+   availability_zone          = (known after apply)
+   cpu_core_count              = (known after apply)
```

6. AWS Output:

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) Running Last updated less than a minute ago Connect Instance state Actions Launch Instance

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic
☒	NewWebServer	i-00e1b045c7d501fb5	Running	t2.micro	2/2 checks passed	View alarms	us-east-2a	-	3.144.14.215	-

i-00e1b045c7d501fb5 (NewWebServer)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance summary Info

Instance ID i-00e1b045c7d501fb5	Public IPv4 address 3.144.14.215 open address	Private IPv4 addresses 10.0.1.198
IPv6 address -	Instance state Running	Public IPv4 DNS -
Hostname type IP name: ip-10-0-1-198.us-east-2.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-1-198.us-east-2.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 3.144.14.215 [Public IP]	VPC ID vpc-097a331d61d1fda8f (MyVPC)	Auto Scaling Group name -
IAM Role -	Subnet ID subnet-0c98bf6c8501558a6 (PublicSubnet)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-2:354918371559:instance/i-00e1b045c7d501fb5	

Security Groups:

i-00e1b045c7d501fb5 (NewWebServer)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Security details

IAM Role -	Owner ID 354918371559	Launch time Fri Feb 28 2025 22:29:14 GMT+0530 (India Standard Time)
Security groups sg-06e8c8601e8866abf (terraform-20250228165901316400000001)		

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-01148851df020e4f2	22	TCP	0.0.0.0/0	terraform-20250228165901316400...	-
-	sgr-03a78e0a31ae8a344	443	TCP	0.0.0.0/0	terraform-20250228165901316400...	-

VPC:

Your VPCs (1/1) Info

Search Last updated less than a minute ago Clear filters

VPC ID : vpc-097a331d61d1fda8f

☒	Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP
☒	MyVPC	vpc-097a331d61d1fda8f	Available	Off	10.0.0.0/16	-	dhcp-0d

vpc-097a331d61d1fda8f / MyVPC

[Details](#) [Resource map](#) [CIDRs](#) [Flow logs](#) [Tags](#) [Integrations](#)

IPv4 CIDRs Info

Address family	CIDR	Status
IPv4	10.0.0.0/16	Associated

Subnet:

subnet-0c98bf6c8501558a6 / PublicSubnet

Details

Subnet ID

subnet-0c98bf6c8501558a6

IPv4 CIDR

10.0.1.0/24

Availability Zone

us-east-2a

Network ACL

acl-035f270642b62afa3

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-2:354918371559:subnet/subnet-0c98bf6c8501558a6

Available IPv4 addresses

250

Availability Zone ID

use2-az1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-097a331d61d1fda8f | MyVPC

Auto-assign public IPv4 address

Yes

Outpost ID

-

Hostname type

IP name

Owner

354918371559

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-0e2490196dde07e01

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Route table: rtb-0e2490196dde07e01

Edit route ta

Routes (1)	
Filter routes	
Destination	Target
10.0.0/16	local